

SANYAM PARESH SHAH

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RESEARCH PROFILE & INTERESTS

Early-career researcher with first author publications in computational health prediction and healthcare informatics and co-authorship in genomics. Demonstrated research independence across Oxford Nanopore sequence analysis, metagenomic analysis, tool development, and machine learning applications, with a track record of contributing to large-scale funded research. Interested in developing rigorous statistical and machine learning methods to analyze genomic data to uncover the genetic and molecular principles underlying human health and disease.

EDUCATION

B.S. Biomedical Informatics and Data Science

December 2025

Arizona State University, Tempe, AZ

Barrett the Honors College

Cumulative GPA: 4.00/4.00 | Dean's List: All Semesters

Relevant Courses:

Comparative Genomics, Functional Genomics, Topics in Translational Bioinformatics, Software Engineering for BMI, Modeling Biomedical Data, Modeling Biomedical Knowledge, Data Structures in Python, Biostatistics, Advanced Topics in BMI, Database Management in Health Care

RESEARCH EXPERIENCE

Undergraduate Research Assistant | Dr Dongwen Wang's Lab | May 2024 – December 2025

- Led recommendation algorithm analysis for \$1.25 million AHRQ-funded project developing an online contraception decision aid for transgender and gender nonconforming individuals assigned female at birth
- Contributed to developing a novel methodology framework to run simulations and conduct sensitivity analyses on the scoring algorithm
- Analyzed scoring algorithm by implementing the novel framework using advanced Python programming, analytical techniques, and 2D/3D visualizations, identifying critical weight thresholds at which recommendations shift and providing insight into the inherent behavior of the algorithm
- Implemented functionality and developed the tool using HTML, CSS and JavaScript.
- Findings resulted in two publications: 1) A full paper in the AMIA 2025 Annual Symposium and 2) an abstract at the Arizona-Nevada Academy of Science, where a poster was presented.

Undergraduate Research Collaborator | August 2024 – December 2025

EMIL Lab Collaboration - Dr. Hassan Ghasemzadeh (Mentor: Abdullah Mamun)

- Developed and implemented MetaBoost, a novel hybrid data balancing method for metabolic syndrome prediction, improving top model performance by 1.7%
- Applied counterfactual analysis to uncover feature importance in mitigating risk for metabolic syndrome and highlighting modifiable factors driving predictions
- Conducted comparative analysis of counterfactual explanation methods (NICE, DiCE, GlyMan) for metabolic syndrome models, analyzing performance through distance, sparsity, and feature modification metrics, identifying trade-offs between each method
- First authored an IEEE EMBC 2025 publication and leading development of an additional first-authored manuscript

Undergraduate Research Assistant | Phan Lab | October 2024 – December 2025

- Systematically benchmarked multiple basecalling models across different trimming parameters on ASU's SOL supercomputer using Dorado software through SLURM job scheduling, finding SUP models produced higher read yields compared to HAC models
- Compared demultiplexing performance across Guppy, Dorado, and PorechopX to evaluate classification efficiency, finding Guppy classified nearly twice as many reads as Dorado
- Developed bash scripts to investigate demultiplexing results and discovered systematic barcode misclassification patterns, particularly involving Barcode 7 in samples where it was never used.

Undergraduate Research Assistant | Dr. Eytayo Owolabi's Lab | August 2025 – December 2025

- Contributed to SmartStart, a NIH-funded K99/R00 project developing a digitally enhanced diabetes self-management intervention for low-income adults newly diagnosed with type 2 diabetes
- Designed a Google Form for data collection to support an early-stage scoping review examining the role of mobile health clinics in addressing transportation barriers to healthcare access for publication
- Conducting literature review on Google Ads utilization for targeted diabetes self-management outreach to reach newly diagnosed patients through tailored DSMES advertising
- Developed a Spanish version of a REDCap survey for a project on brain health literacy among adults with type 2 diabetes, comprising 20+ sections with branching logic, automated routing, and file uploads to enable inclusive data collection
- Learned qualitative research methodologies including interview guide design, interviewing, data management, analysis and report writing

Undergraduate Research Assistant | Gevork Harootunian, AACOM | August 2025 – November 2025

- Analyzed survey data from graduating Doctor of Osteopathic Medicine students to examine relationships between perceived bias, burnout, resilience, belonging, and overall well-being
- Conducted response rate analysis and comprehensive data preprocessing for multi-institutional medical education research
- Wrote a fully annotated Python analysis script to assess potential non-response bias in survey data through demographic comparisons, chi-square and ANOVA testing, and data visualization

PUBLICATIONS

1. **Shah S.P**, Shekhawat T.S, Hoang V-A, Chiou E, Brian J, Wang D. Sensitivity Analyses of a Scoring System for a Contraception Decision Aid. *AMIA Annu Symp Proc.* 2025 May 22;2024:1150-1159. PMID: 41726543; PMCID: PMC12919583. **[Full Paper, Peer-Reviewed Conference]**
2. **Shah, S.P.**, Mamun, A., Soumma, S.B., Ghasemzadeh, H (2025). Enhancing Metabolic Syndrome Prediction with Hybrid Data Balancing and Counterfactuals. *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Copenhagen, Denmark, 2025, pp. 1-7, doi: 10.1109/EMBC58623.2025.11253729. **[Full Paper, Peer-Reviewed Conference]**
3. Agoti, CN., Njeru, R., Mwanga, M., Mutunga, M., **Shah, S.P.**, Chu, T., Nokes, DJ., Cotten, M., Phan, MVT. (2025). Cryptosporidium parvum Virus 1 Genome Sequences from Pediatric Diarrhea Cases in Coastal Kenya. *Microbiology Resource Announcements*, 14(3), e00349-25. <https://doi.org/10.1128/mra.00349-25> **[Peer-Reviewed Journal]**
4. **Shah, S.P.**, Mamun, A., Soumma, S.B., Ghasemzadeh, H. MetaBoost: Hybrid Data Balancing and Counterfactual Explanations for Early and Interpretable Prediction of Metabolic Syndrome. *(Manuscript in preparation)*

PRESENTATIONS

1. **Shah, S.P.**, Shekhawat, T.S., Hoang, V.A., Chiou, E., Brian, J., Wang, D. (2025). "Sensitivity Analyses of a Scoring System for an Online Contraception Decision Aid."
 1. *Arizona-Nevada Academy of Science Annual Meeting*, Phoenix, AZ, 2025 (Poster) [Proceedings of the Arizona-Nevada Academy of Science, Vol. 57, p. 48 (2025)]
 2. *College of Health Solutions Student Research Symposium*, Phoenix, AZ, 2025 (Poster)
 3. *AMIA Annual Symposium*, Atlanta, GA, 2025 (Oral presentation)
2. Sneddon, J., Sithemsetti, N., **Shah, S.P.**, Ruiz, F.R. (2025). "Improving genetic testing for cancer treatment using the Toyota Production System A3 Methodology."
 1. *College of Health Solutions Student Research Symposium*, Phoenix, AZ, 2025 (Poster)
 2. *BMI 404/504 Mock Boards Pitch Presentation*, Phoenix, AZ, 2025 (Oral Presentation) [**First Place Award**]

HONORS AND AWARDS

Awards

- **University Graduate Fellowship** – Penn State University (2026)
- **Barrett Project Fund** - \$1,000 for AMIA 2025 Annual Symposium presentation (2025)
- **CHS Student Conference Support** - \$500 for AMIA 2025 Annual Symposium presentation (2025)
- **Barrett Thesis Fund** - \$260 for honors thesis poetry collection publication (2025)

Academic Honors

- **Moeur Award** – Arizona State University (2025)
- **Dean's List** – All Semesters (2022-Present)
- **Barrett Honors College Scholar** – Arizona State University
- **First Place Certificate** – BMI 404/504 Mock Boards Pitch Presentation (2025)
- **SUN Award** – Arizona State University (2024, 2025)

TEACHING AND PROFESSIONAL EXPERIENCE

Teaching Assistant | BMI 465: Comparative Genomics | August 2025 – December 2025

- Evaluated student participation and engagement in course activities
- Delivered a 25-minute introduction to Geneious software for genomic analysis
- Providing weekly hands-on support during practical sessions and assisting students with bioinformatics software and data analysis workflows

Teaching Assistant | HCD 300: Biostatistics | August 2024 – December 2024

- Conducted regular office hours to provide academic support and clarify course material
- Graded Excel-based assignments and provided constructive feedback on data analysis techniques
- Facilitated online discussion boards and provided timely responses to student inquiries

Teaching Assistant | BMI 110: Introduction to Python | January 2024 – May 2024

- Led weekly recitation sessions, developing hands-on Python exercises to reinforce core programming concepts
- Graded programming assignments and provided detailed feedback on syntax, logic, and code structure
- Conducted office hours and assisted students with debugging, troubleshooting code errors, and resolving programming issues
- Assisted with lecture delivery, managed live coding demonstrations, and delivered a full lecture on object-oriented programming

Subject Area Tutor | Arizona State University | November 2024 – December 2025

- Delivered individualized tutoring for Biology, Statistics, Chemistry, Mathematics, Python, SQL, R, Bash Scripting, Genetics and Python
- Coordinated group tutoring sessions, managing multiple students simultaneously while ensuring each receives targeted support and maintains engagement
- Created and posted educational materials on Inscribe platform and provided timely responses to student questions and inquiries

Barrett Peer Mentor | Arizona State University | August 2024 – December 2025

- Mentored 100+ students on academics, campus resources, and honors program navigation
- Organized monthly community-building events and educational workshops, delivered presentations and prepared necessary materials to foster engagement and academic support
- Collaborated with peer academic leaders and other ASU organizations to coordinate large-scale programming initiatives
- Sent weekly event notifications and GroupMe announcements while providing ongoing support by answering student questions and concerns

ACADEMIC PROJECTS

Health Risk Prediction Website

- Developed a web-based tool integrating multiple machine learning and deep learning models to predict health risks, including metabolic syndrome, stroke, and breast cancer
- Enabled users to select a condition, input personal health data, and receive model-generated predictions via email.
- Implementing counterfactual analysis and AI-powered explanations to provide users with personalized risk mitigation strategies and actionable intervention steps

Cancer Prediction Using Deep Learning

- Collaborated with a local hospital to develop a convolutional neural network for classifying breast ultrasound images
- Successfully predicted the presence of cancer based on imaging data achieving an accuracy of 94%
- GitHub: [cancer_prediction_cnn](#)

TECHNICAL SKILLS

- **Programming Languages:** Python, R, C++, SQL, Bash, Git, LaTeX
- **Bioinformatics Tools:** Guppy, Dorado, Porechop, BLAST, MMseqs2, SAMtools, MAFFT, FastQC, Trimmomatic
- **Workflow & HPC:** SLURM, Docker, ASU SOL supercomputer
- **Machine Learning:** Random Forest, SVM, CNNs, XGBoost, SHAP, clustering, Counterfactuals
- **Data Analysis:** scipy, statsmodels, scikit-learn hypothesis testing, data visualization, pandas, NumPy
- **Development:** Web development, GUI development, Database design, Redcap
- **Data Visualization:** ggplot2, pyplot (3D Visualization), seaborn, matplotlib

SCIENCE COMMUNICATION & OUTREACH

1. **Author, *Bioinformatics*** (Self-Published, 2025)
A collection of 15 scientific poems exploring areas such as genomics, proteomics, and bioinformatics. Poems also featured in [Canyon Voices Literary Journal](#) and [ChART Literary Journal](#) (Peer-Reviewed). [\[Poetry Link\]](#)